

Bringing Metropolis–Hastings Sampling to NuWro

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September 18, 2026

Section 1

Problem and physics objectives

Motivation for event-by-event MCMC

- The current workflow estimates each channel's integrated cross section and maximum weight, then generates equal-weight events channel by channel.
- A new maximum weight may require reducing earlier samples. Events already passed to downstream simulation are difficult to retract.¹
- MH needs no predetermined global weight bound and suits streaming output.
- The costs are chain correlations, initial transients, and a more demanding assessment of effective statistics.

¹This is related to the JUNO framework, which does not allow pre-generating to a ROOT file and reading from it.

Cross-section integrals and event weights

For channel c , map the kinematic variables to a unit hypercube $\mathbf{u} \in [0, 1]^{d_c}$:

$$F_c(\mathbf{u}) \equiv \frac{d\sigma_c}{d^{d_c}\mathbf{u}} = \left. \frac{d\sigma_c}{d^{d_c}\boldsymbol{\xi}} \right|_{\boldsymbol{\xi}=\boldsymbol{\xi}_c(\mathbf{u})} \left| \det \frac{\partial \boldsymbol{\xi}_c}{\partial \mathbf{u}} \right|, \quad \sigma_c = \int F_c(\mathbf{u}) d^{d_c}\mathbf{u}.$$

- F_c includes the Jacobian and has cross-section units. For n_c uniform proposal candidates, $\hat{\sigma}_c = n_c^{-1} \sum_m F_c(\mathbf{u}_m)$.
- Weighted events: proposal weights estimate integrals and histograms.
- Equal-weight events: samples follow the multichannel mixture F_c/σ_{tot} , with the total cross section stored separately.

$$\sigma_{\text{tot}} = \sum_c \sigma_c, \quad \left. \frac{d\sigma}{dz} \right|_{B_i} \simeq \frac{\hat{\sigma}_{\text{tot}}}{N \Delta z_i} \sum_{n=1}^N \mathbf{1}[z_n \in B_i].$$

Rejection sampling: efficiency and maximum weight

With uniform proposals on the unit hypercube, the acceptance probability in channel c is

$$a_{\text{AR},c}(\mathbf{u}) = \frac{F_c(\mathbf{u})}{F_{c,\text{max}}}, \quad \epsilon_{\text{AR},c} = \frac{\sigma_c}{F_{c,\text{max}}}.$$

- Test runs estimate σ_c and $F_{c,\text{max}}$. Channel event counts follow the cross sections.
- If the bound rises from $F_{c,\text{max}}^{\text{old}}$ to $F_{c,\text{max}}^{\text{new}}$, earlier accepted counts are reduced by $F_{c,\text{max}}^{\text{old}}/F_{c,\text{max}}^{\text{new}}$: as if they were sampled using the new bound at the time of acceptance.
- Extreme weights in a small phase space can suppress acceptance throughout a channel.
- Better variable transformations and importance proposals (e.g. dismode) can also improve rejection sampling. MH offers another option.

Multichannel phase space and importance weights

$$\mathcal{X} = \bigsqcup_c \{c\} \times [0, 1]^{d_c}, \quad \pi(c, \mathbf{u}) = \frac{F_c(\mathbf{u})}{\sigma_{\text{tot}}}, \quad \sigma_{\text{tot}} = \sum_c \int F_c(\mathbf{u}) \, \mathrm{d}^{d_c} \mathbf{u}.$$

Select channel c with probability p_c^{prop} , then draw uniformly within its unit hypercube:

$$q(c, \mathbf{u}) = p_c^{\text{prop}}, \quad w_{\text{IS}}(c, \mathbf{u}) = \frac{F_c(\mathbf{u})}{p_c^{\text{prop}}}, \quad \mathbb{E}_q[w_{\text{IS}}] = \sigma_{\text{tot}}.$$

$$\langle O \rangle_\pi = \frac{\mathbb{E}_q[w_{\text{IS}} O]}{\mathbb{E}_q[w_{\text{IS}}]} \simeq \frac{\sum_m w_{\text{IS},m} O_m}{\sum_m w_{\text{IS},m}}.$$

Section 2

Algorithm and correctness

The Metropolis–Hastings Algorithm

At current state x , propose a candidate y from $q(y \mid x)$:

$$\alpha(y \mid x) = \min \left\{ 1, \frac{\pi(y)q(x \mid y)}{\pi(x)q(y \mid x)} \right\}.$$

1. Draw $U \sim \text{Uniform}(0, 1)$.
2. If $U < \alpha$, move to y . Otherwise remain at x .
3. **A rejection is also a chain iteration.** Repeated residence at the current state must enter the sample sequence.

$$\pi(x)q(y \mid x)\alpha(y \mid x) = \min\{\pi(x)q(y \mid x), \pi(y)q(x \mid y)\}.$$

The right-hand side is unchanged by exchanging x, y , giving off-diagonal detailed balance.

The full transition kernel and its conditions

Including residence after rejection, the transition kernel is

$$T(x, dy) = q(y | x)\alpha(y | x) dy + r(x)\delta_x(dy),$$

$$r(x) = 1 - \int q(y | x)\alpha(y | x) dy.$$

- Detailed balance gives invariance. Convergence also requires suitable irreducibility, aperiodicity, and related conditions.
- q must cover the target support. Regions with non-zero cross section must remain reachable by proposals.
- Channels differ in dimension and have no natural neighbouring relation, so a local proposal across channels is ill-defined. Independent proposals on the disjoint union avoid the problem by needing no shared coordinates.

Independent proposals: reusing NuWro's makeevent

With $q(y | x) = q(y)$,

$$\alpha(y | x) = \min \left\{ 1, \frac{w_{\text{IS}}(y)}{w_{\text{IS}}(x)} \right\}, \quad w_{\text{IS}}(x) = \frac{F(x)}{q(x)}, \quad F(x) = \sigma_{\text{tot}} \pi(x).$$

Thus $\pi(x)/q(x) = w_{\text{IS}}(x)/\sigma_{\text{tot}}$; the cross-section normalization cancels in α .

- Reuse each physics model's existing generation of degrees of freedom, without constructing local phase-space steps.
- Candidates do not depend on the current point. The proposal has no local random walk.
- **Independent candidates do not imply independent outputs:** rejections create repeated states and correlations.
- Across channels, q is generally not symmetric under exchanging x and y (biased proposal). The p_c^{prop} factors cannot be dropped unconditionally.

Energy spectra and dismode: importance factors

Normalize the flux ϕ over the selected energy range:

$$f_\nu(E_\nu) = \frac{\phi(E_\nu)}{\int \phi(E_\nu) dE_\nu}, \quad \bar{\sigma}_c \equiv \langle \sigma_c \rangle_{f_\nu} = \int f_\nu(E_\nu) \sigma_c(E_\nu) dE_\nu.$$

For energy proposal q_E and uniform $\mathbf{u}$ within channel c ,

$$w_{\text{int}} = \frac{f_\nu(E_\nu) F_c(E_\nu, \mathbf{u})}{q_E(E_\nu)}, \quad w_{\text{IS}} = \frac{w_{\text{int}}}{p_c^{\text{prop}}}, \quad \mathbb{E}_q[w_{\text{IS}}] = \bar{\sigma}_{\text{tot}} \equiv \sum_c \bar{\sigma}_c.$$

For a spectrum, use $\bar{\sigma}_c$ and $\bar{\sigma}_{\text{tot}}$ in place of σ_c and σ_{tot} throughout.

- w_{raw} denotes the NuWro weight before beam-proposal correction. For $q_E = f_\nu$, $w_{\text{int}} = w_{\text{raw}}$ if all other factors are included.
- For $q_E = E_\nu f_\nu / \langle E_\nu \rangle_{f_\nu}$ and no prior energy-bias correction, $w_{\text{int}} = \langle E_\nu \rangle_{f_\nu} w_{\text{raw}} / E_\nu$. The common mean energy cancels only in MH ratios.
- MH enables `dismode=1` independently of DIS/RES selection. For a monoenergetic beam, the common energy cancels in the ratio.

Uniform and weighted channel selection

Suppose C reaction channels are enabled.

$$\text{Uniform: } p_c^{\text{prop}} = 1/C, \quad \alpha = \min\{1, w_{\text{int}}(y)/w_{\text{int}}(x)\};$$

$$\text{Weighted: } p_c^{\text{prop}} \neq p_{c'}^{\text{prop}}, \quad \alpha = \min\left\{1, \frac{w_{\text{int}}(y)/p_{c'}^{\text{prop}}}{w_{\text{int}}(x)/p_c^{\text{prop}}}\right\}.$$

Here $c = c(x)$, $c' = c(y)$; w_{int} includes the beam-proposal correction.

- Uniform selection wastes many candidates when channel cross sections differ greatly.
- $p_c^{\text{prop}} \propto \sigma_c$ compensates for average scale differences **between channels**.
- Even optimal channel fractions leave within-channel peaks in w_{IS} . Acceptance need not approach 1.
- Output channel frequencies follow $P_c = \sigma_c/\sigma_{\text{tot}}$, not the proposal probabilities p_c^{prop} .

Cross-section estimates require proposal samples

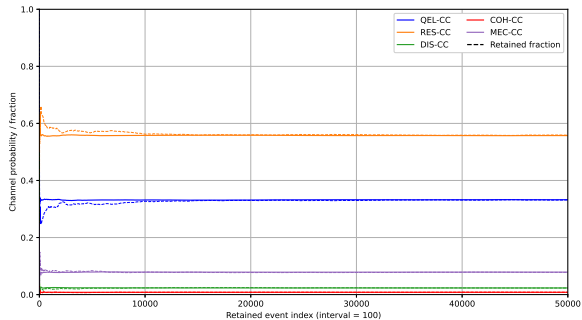
The sources estimate cross sections from each channel's `makeevent` candidates, including those rejected by MH:

$$\hat{\sigma}_c = \frac{1}{n_c} \sum_{m: c_m=c} w_{\text{int},m}, \quad p_c^{\text{prop}} = \frac{\hat{\sigma}_c}{\sum_j \hat{\sigma}_j}.$$

w_{int} is fully corrected for the beam and within-channel proposal; $w_{\text{IS}} = w_{\text{int}}/p_c^{\text{prop}}$ also corrects channel selection.

- n_c counts all candidates in channel c . Valid zero-weight candidates count in n_c ; omitting them overestimates the cross section.
- Start with uniform selection and update p_c^{prop} after each retained sample.

Evolution of channel proposal probabilities



^{12}C , 2 GeV, all CC, retention interval 100. Solid lines: proposal probabilities. Dashed lines: cumulative channel fractions among retained events (including repeated states).

Fixed-interval retention and burn-in

To further reduce correlations, the states of the MCMC chain are thinned by a fixed interval.

$$Y_n = X_{nk_{\text{thin}}}, \quad n = 1, \dots, N.$$

- k_{thin} : retention interval, reducing correlations between adjacent stored events.
- For a fixed kernel T , Y has transition kernel $T^{k_{\text{thin}}}$ and the same invariant distribution π .
- The cost of repeating sampling is acceptable because FSI is only applied to the retained states.

Section 3

Implementation and interfaces

Implementation sequence for event-by-event output

```
initialize positive valid pre-FSI state x; p_prop[c] = 1/C
for each requested output event:
    repeat k_thin = mh_sample_interval times: // p_prop fixed
        c = sample_channel(p_prop)
        y = makeevent(c)                        // pre-FSI candidate
        update_channel_integral(y)             // before MH selection
        w_IS_y = integration_weight(y) / p_prop[c]
        w_IS_x = integration_weight(x) / p_prop[channel(x)]
        if uniform01() < min(1, w_IS_y/w_IS_x):
            x = y                               // otherwise retain x
    out = copy(x)
    finishevent(out); save(out)                // FSI only for output
    update p_prop if adaptation is enabled
```

Expensive FSI after sampling

Pre-FSI candidate $\longrightarrow$ MH + fixed interval $\longrightarrow$ FSI $\longrightarrow$ Save.

- Candidates require only primary interactions and weights. Each actual output invokes FSI once.
- Increasing k_{thin} mainly adds proposal cost. It does not send every candidate through the intranuclear cascade.
- Repeating a pre-FSI state can yield different final states when FSI is resampled.
- FSI randomness cannot remove all primary-kinematic correlations. Check both pre- and post-FSI observables.
- Isolate the chain state from the mutable FSI event object before saving, to preserve the next acceptance ratio.

Parameters and exceptional states in the sources

Source parameter	Meaning and status
<code>use_mh</code>	MH switch. Default to false (opt-in)
<code>mh_sample_interval</code>	Proposals between outputs. Must be a positive integer.
<code>use_weighted_channel</code>	Adaptive channel-weighting switch, labeled <code>wc</code> in plots.
<code>number_of_test_events</code>	Ignored in MH mode.
<code>random_seed</code>	Shared MT19937, keeps reproducibility with MH.

Section 4

Correlations and statistics

Repeated events must remain in the sample

- A state with large $w_{IS}(x)$ may reject candidates for a long time. Its residence time contributes to the target probability mass.
- Removing duplicates or saving only accepted proposals produces a biased jump-chain sample.
- Save the state every k_{thin} steps on a fixed clock, allowing repeated occurrences.

Autocorrelation is observable-specific

For $O_n = O(Y_n)$ on the retained sequence, use

$$\hat{\gamma}_O(\ell) = \frac{1}{N-\ell} \sum_{n=1}^{N-\ell} (O_n - \bar{O})(O_{n+\ell} - \bar{O}), \quad \hat{\rho}_O(\ell) = \frac{\hat{\gamma}_O(\ell)}{\hat{\gamma}_O(0)}.$$

- The current benchmark uses a scalar event weight of the sampled event.
- Under a white-noise null, $\pm 1.96/\sqrt{N}$ is an approximate pointwise reference band, not a simultaneous band over all lags.²

²A tricky part is with weighted channel, the MCMC chain is adaptive and the retained sequence is not strictly stationary. But such comparison still make sense if considered qualitatively not quantitatively.

Integrated autocorrelation time and ESS

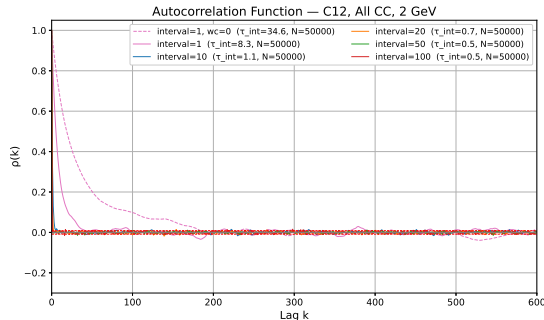
Throughout, use the original paper's half-value convention:

$$\tau_{\text{int},O} = \frac{1}{2} + \sum_{\ell=1}^{\infty} \rho_O(\ell), \quad g_O = 2\tau_{\text{int},O}, \quad N_{\text{eff},O} = \frac{N}{g_O}.$$

$$\text{Var}(\bar{O}) \simeq \frac{\text{Var}_{\pi}(O)}{N} g_O.$$

- Independent samples: $\tau_{\text{int},O} = 1/2$, $g_O = 1$, $N_{\text{eff},O} = N$.
- If $\tau_{\text{int},O} = 10$, one effective sample requires about 20 **samples of the analyzed sequence**.

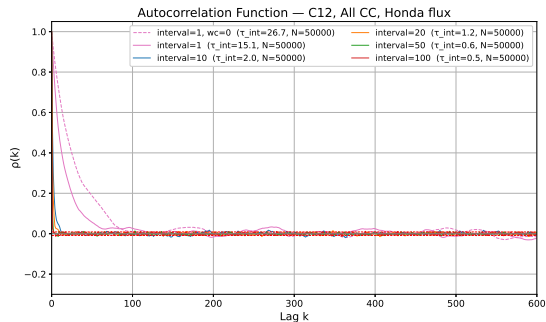
Autocorrelation with a monoenergetic beam



^{12}C , 2 GeV, all CC. “interval” is the retention interval. wc=0 disables channel weighting.

- For interval=1, weighted channel improves the autocorrelation significantly
- A large interval removes long-residence correlations. Interval of 50 yields near independent samples.

Autocorrelation with the Honda spectrum



^{12}C , Honda flux, all CC. The sources report small correlations in the measured weight at interval=100.

A wide-spread beam still gives good autocorrelation at interval=100.

Parallel chains: bounding standard-error inflation

^{12}C , $E_\nu = 2$ GeV, CC without COH; FSI on, frozen channel proposal.

Method: parallel chains + batch means

- Four replicas with independent seeds: 4×48 chains per interval k .
- Split each retained chain into batches of b events; pool batch moments across chains.

$$\hat{g}_h = \frac{b s_{\text{batch means},h}^2}{s_h^2}, \quad \Delta_h = 100(\sqrt{\hat{g}_h} - 1)\%.$$

Δ_h : increase in the **standard error of a mean or bin frequency**, relative to IID events. Check batch-size stability, residual batch dependence and rank- $\hat{R}$.

Results across physics observables

Retention interval k	20	100
Complete chains	192	191
Retained events	173.02M	86.06M
Batch size b	4096	2048
Support $\Delta_h \leq 5\%$	19/22	22/22

$k = 100$: all 22 estimable observables

Standard-error increase $< 2.15\%$

at 95% simultaneous asymptotic confidence*.

Channels, Q^2/W bins and tails, CC- 0π , pion multiplicity and nucleon KE.

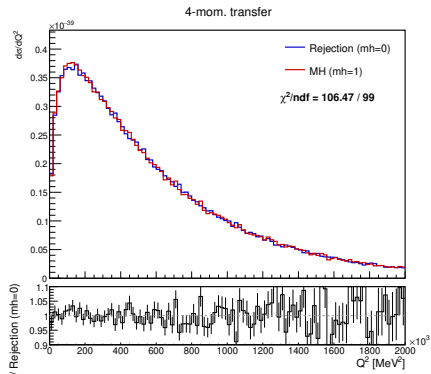
$k = 20$: two low- Q^2 bins inconclusive; one W bin fails the batch check.

* Joint batch-moment intervals; Bonferroni over 48 endpoints and two looks (familywise target $\geq 95\%$). Assumes stationary mixing; no finite-sample guarantee. Two constant endpoints per k are not estimable.

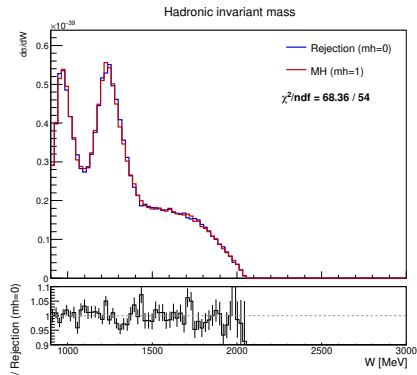
Section 5

Validation and performance

Differential cross sections: 2 GeV, all CC



$^{12}\text{C}, d\sigma/dQ^2$



$^{12}\text{C}, d\sigma/dW.$

The sources report agreement between MH and rejection sampling.

MINERvA TKI: validation at the final-state level

- Updated NuWro (a67f0e7): hybrid + LFG, identical physics for MH and rejection; comparison plots in backup.
- RES-DIS transition: 1.6–1.9 GeV, cut 1.9 GeV. Each sampler: 25.6M events; MH retention interval 80.

χ^2 against data	ndf	Rejection	MH
CC- 0π , $\delta\alpha_T$	12	22.64	23.72
CC- 0π , p_N	24	78.54	76.86
CC- π^0 , $\delta\alpha_T$	9	4.07	4.12
CC- π^0 , p_N	12	137.44	139.00
CC- π^0 , δp_{TT}	13	33.48	33.97

Full experimental covariance; no fitted parameters. Carbon-only CH normalization: 12/13.

Section 6

Further proposal optimization

Ideal independent proposals: flatter target/proposal ratios

$$\alpha(y \mid x) = \min \left\{ 1, \frac{\pi(y)/q(y)}{\pi(x)/q(x)} \right\}.$$

If $q = \pi$, then $\alpha = 1$. A proposal closer to the target usually stabilizes the acceptance ratio.

- Channel weighting learns only a low-dimensional discrete mixture fraction.
- A possible improvement is to collect per-channel phase-space information.
- Retain independent proposals to avoid overly narrow local random walks.

Adaptive histograms within each channel

- A possible approach: several histograms for each channel's internal degrees of freedom to learn proposals online.
- Specify a computable, $q_c(\mathbf{u})$ and include it in the full MH acceptance ratio.
- Similar criteria and logic as adaptive channel weighting: burn-in, frozen proposal, and convergence diagnostics.

A **supplementary design** that covers the support:

$$q_{\text{mix}}(x) = (1 - \epsilon)q_{\text{learned}}(x) + \epsilon q_0(x), \quad 0 < \epsilon < 1,$$

Here q_0 is positive on the target support. Acceptance uses the full mixture density.

Planned future work.

As a gateway to Normalizing Flow

- Normalizing Flow already provides a differentiable, invertible map from a simple base distribution to a complex target distribution.
- With rejection sampling, one still have to estimate the maximum of the target/proposal ratio, which is difficult in high dimensions.
- MH would provide a valuable alternative to rejection sampling, when working with a surrogate proposal distribution.

Planned future work.

Available evidence and remaining work

- Current MCMC based sampling implementation is validated against the original rejection sampling in terms of cross section and shape.
- Techniques such as thinning and adaptive channel-proposal weighting provides good autocorrelation suppression.
- The comparison suggests that the MCMC based sampling produce statistically equivalent results to the original rejection sampling.
- MCMC would open a gateway for more advanced proposal optimization.

Thank you!

$\delta\alpha_T$ measurements

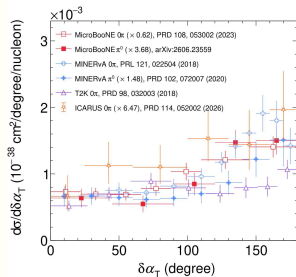


Figure 1 The numerical inputs are the published MicroBooNE, MINERvA, and T2K releases, the digitized MicroBooNE π^0 result, and the ICARUS HEPData values. Scale factors shown in the legend are display choices applied for a common shape comparison; they are not additional measurements. Sources [1], [2], [3], [4], [5], and [6].
Download PDF / EPS

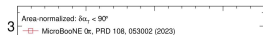
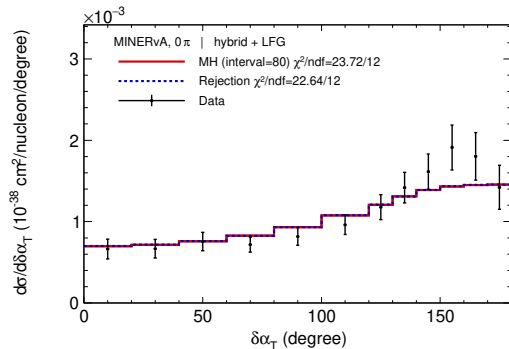


Figure 2 Every distribution is scaled to a common interpolated integral over $0 \leq \delta\alpha_T < 90$ degrees, so the vertical axis is the differential shape normalized by the partial integral $\int_0^{\delta\alpha_T} d\alpha_T$. The interpolation uses the bin-center histograms and is linear within each bin; for ICARUS, the companion histogram supplies the bin

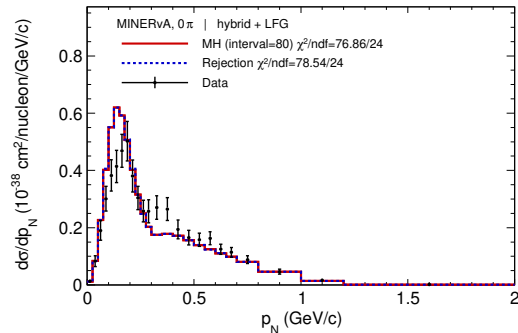
Just an advertisement:
A Compilation of TKI Cross
Sections Figures:
<https://tki.yanqiyu.info/>
Will update routinely.

backup

MINERvA TKI: CC- 0π



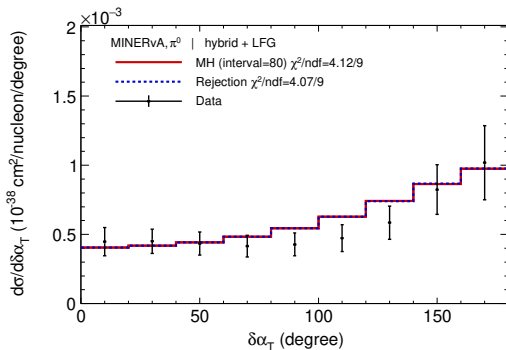
$\delta\alpha_T$



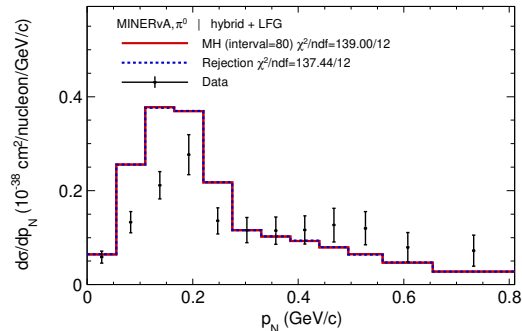
Reconstructed nucleon momentum p_N

Hybrid + LFG; 25.6M events per sampler; MH interval 80. Full-covariance χ^2/ndf in legends; data bars show diagonal errors.

MINERvA TKI: CC- π^0



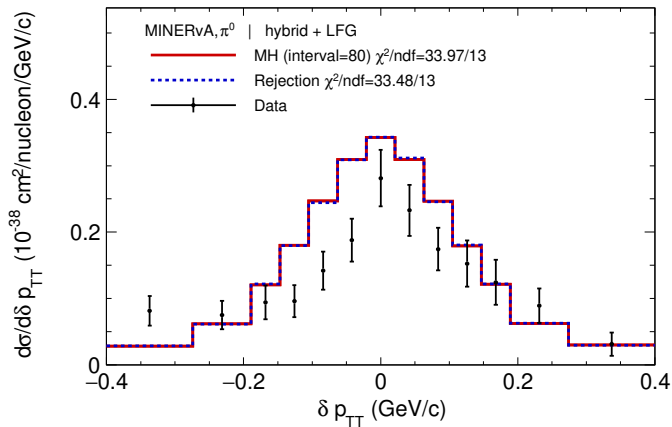
$\delta\alpha_T$



Reconstructed nucleon momentum p_N

Hybrid + LFG; RES-DIS transition 1.6–1.9 GeV, cut 1.9 GeV. Identical physics and normalization for MH and rejection.

MINERvA TKI: $\text{CC-}\pi^0$, δp_{TT}



Hybrid + LFG; 25.6M events per sampler; MH interval 80.

Appendix: detailed balance under fixed-interval retention

Write $\pi(dx) = \pi(x) dx$ for the target probability measure. For the full, fixed, reversible kernel,

$$\pi(dx) T(x, dy) = \pi(dy) T(y, dx).$$

Repeated detailed balance reverses the stationary path measure:

$$\pi(dx_0) \prod_{j=0}^{k_{\text{thin}}-1} T(x_j, dx_{j+1}) = \pi(dx_{k_{\text{thin}}}) \prod_{j=0}^{k_{\text{thin}}-1} T(x_{j+1}, dx_j).$$

Integrating out all intermediate states gives

$$\pi(dx) T^{k_{\text{thin}}}(x, dy) = \pi(dy) T^{k_{\text{thin}}}(y, dx).$$

- Kernel measures include the rejection atom and mixed discrete/continuous states.
- $T^{k_{\text{thin}}}$ preserves π . The proof neither requires nor guarantees independent retained samples.

Appendix: absolute integrals versus MH chain averages

For a fixed proposal q , with $F = \sigma_{\text{tot}}\pi$ and $w_{\text{IS}} = F/q$:

$$\mathbb{E}_q[w_{\text{IS}}] = \int F(x) dx = \sigma_{\text{tot}},$$

$$\mathbb{E}_\pi[w_{\text{IS}}] = \frac{1}{\sigma_{\text{tot}}} \int \frac{F(x)^2}{q(x)} dx \neq \sigma_{\text{tot}} \quad \text{in general.}$$

- Estimate the total cross section using a defined proposal measure. The chain generates equal-weight events with the correct shape.
- At step t , draw candidate y_t from q_t and set $w_{\text{IS},t} = F(y_t)/q_t(y_t)$. If $q_t > 0$ on the target support, then $\mathbb{E}[w_{\text{IS},t} | \mathcal{F}_{t-1}] = \sigma_{\text{tot}}$, where $\mathcal{F}_{t-1}$ is the sampling history.
- Record flux normalization, integration range, target composition, and units together with the integration weights.

Appendix: physical meaning of TKI variables

Let ℓ be the charged lepton and N the selected hadronic system:

$$\delta\vec{p}_T = \vec{p}_T^\ell + \vec{p}_T^N, \quad \delta\alpha_T = \arccos \frac{-\vec{p}_T^\ell \cdot \delta\vec{p}_T}{|\vec{p}_T^\ell| |\delta\vec{p}_T|}.$$

$$R_A = M_A + p_L^\ell + p_L^N - E^\ell - E^N, \quad \delta p_L = \frac{R_A^2 - |\delta\vec{p}_T|^2 - (M_{A-1}^*)^2}{2R_A},$$

$$p_n = \sqrt{|\delta\vec{p}_T|^2 + \delta p_L^2}, \quad M_{A-1}^* = M_A - M_n + \epsilon_{\text{rem}}.$$

ϵ_{rem} is the effective neutron removal energy in the residual-mass prescription; S6 uses $\epsilon_{\text{rem}} = 28.7$ MeV for C12. CC-0 π selects a proton. CC- π^0 uses the corresponding $p + \pi$ system. Consult the original analyses for detailed selection cuts.

Window estimates and finite-chain diagnostics

In practice, the calculation needs a finite sum. The Sokal windowing method uses:

$$\hat{\tau}_{\text{int},O}(L) = \frac{1}{2} + \sum_{\ell=1}^L \hat{\rho}_O(\ell),$$

$$L_{\text{win}} = \min\{L \geq 1 : L \geq C_{\text{win}} \hat{\tau}_{\text{int},O}(L)\}, \quad C_{\text{win}} = 5.$$

- Purpose: prevent noisy large-lag estimates from dominating the infinite sum.
- Report N , k_{thin} , burn-in, L_{win} , and the convention for $\hat{\tau}_{\text{int},O}$.
- For $\hat{g}_O(L) = 1 + 2 \sum_{\ell=1}^L \hat{\rho}_O(\ell)$, the condition is $L_{\text{win}} \geq (C_{\text{win}}/2) \hat{g}_O(L_{\text{win}})$.
- If no stable window exists, the chain is too short, or nonstationarity is evident, extend the run and inspect traces.
- Compare means, channel fractions, rare regions, and block errors across initial states and random seeds.

Historical performance: 50,000 Honda/JUNO events

Method	Runtime/s	Source unique rate
MH, $k_{\text{thin}} = 20$	21.20	63.09%
MH, $k_{\text{thin}} = 100$	86.22	99.96%
Rejection, 15,000,000 test 209 (test)+137 (real)		Not given

- For this test, the total runtime ratio at $k_{\text{thin}} = 100$ is $(209 + 137)/86.22 \simeq 4.01$.
- Comparing only the real run gives $137/86.22 \simeq 1.59$. State whether startup costs are amortized.
- $k_{\text{thin}} = 20$ is fast but has substantial duplication. Runtime ratios alone do not measure gains in independent statistics.

Extreme weights: Ghent/PYTHIA and SPPF

A special case: in the Hybrid implementation, the weight can peak at the Δ region.

- Rare regions produce very large weight bounds.
- MH avoids direct global suppression by this bound, but can remain at large-weight states for a long time.
- The global-bound problem becomes a mixing problem. MH is less sensitive to the peak.